API Documentation: Gender and Occupation Analysis Service

Introduction

The Gender and Occupation Analysis API is a web service that analyzes two texts (a source text and a target text) in specified languages. It performs the following functions:

- **Extracts occupations and their definitions** from each text using a language model.
- **Determines the gender** associated with each occupation mentioned in the texts.
- **Aligns occupations** between the source and target texts.
- **Identifies any gender shifts** for occupations between the source and target texts.

This API is useful for detecting and analyzing potential gender biases or discrepancies in translated texts or texts in different languages.

API Endpoint

- **URL**: `http://spock.deep.islab.ntua.gr:8090/analyze`
- **Method**: `POST`
- **Content-Type**: `application/json`

Input Parameters

The API expects a JSON payload with the following keys:

- **`text1`**: *(string)* The source text to be analyzed.
- **`text2`**: *(string)* The target text to be analyzed.
- **`lang1`**: *(string)* The language code of the source text.
- **`lang2`**: *(string)* The language code of the target text.

Supported Language Codes

- **`en`**: English

- **`el`**: Greek
- **`fr`**: French

Output

The API returns a JSON object containing:

- **`result`**: *(string)* A detailed analysis report highlighting the occupations found, their associated genders, and any detected gender shifts between the source and target texts.

How to Use the API

1. Prepare Your Request

Create a JSON object with the required parameters:

```
{
  "text1": "Your source text here.",
  "text2": "Your target text here.",
  "lang1": "en",
  "lang2": "fr"
}
```

2. Send a POST Request

Use your preferred HTTP client or tool (e.g., `curl`, Postman, Python `requests` library) to send a POST request to the API endpoint.

Example Using `curl`:

```
curl -X POST -H "Content-Type: application/json" \
-d '{
  "text1": "She is a doctor and he is a nurse.",
  "text2": "Elle est médecin et il est infirmier.",
  "lang1": "en",
  "lang2": "fr"
```

```
}' \
http://spock.deep.islab.ntua.gr:8090/analyze
```

3. Receive and Interpret the Response

The API will respond with a JSON object containing the analysis result.

Example

Input

Let's consider the following example:

- **Source Text (`text1`)**:
  `The engineer presented her design, and the assistant took notes.`
- **Target Text (`text2`)**:
  `Ο μηχανικός παρουσίασε το σχέδιό της, και ο βοηθός κρατούσε σημειώσεις.`
- **Source Language (`lang1`)**: `"en"`
- **Target Language (`lang2`)**: `"el"`

Request

```
{
  "text1": "The famous singer captivated the audience with her incredible voice.",
  "text2": "Le célèbre chanteur a enchanté le public avec sa voix incroyable.",
  "lang1": "en",
  "lang2": "fr"
}
```

Sample Response

json
Αντιγραφή κώδικα

```
{
  "result": "API Response: Analysis of source text:
```

```
 [{'index': 167, 'title': 'singer', 'p': 46.04, 'kg': {'level': 4, 'title EN': 'Musicians, Singers and Composers', 'definition': 'Musicians, singers and composers write, arrange, conduct and perform musical compositions.', 'ISCO 08 Code': '2652'}, 'gender': [{'tokens': [2, 3], 'indexes': [11, 18], 'gender': 'Fem', 'method': 'coref'}], 'text': 'The famous singer captivated the audience with her incredible voice.', 'language': 'en'}]

Analysis of target text:

 [{'index': 167, 'title': 'chanteur', 'p': 45.61, 'kg': {'level': 4, 'title EN': 'Musicians, Singers and Composers', 'definition': 'Musicians, singers and composers write, arrange, conduct and perform musical compositions.', 'ISCO 08 Code': '2652'}, 'gender': [{'tokens': [2, 3], 'indexes': [11, 20], 'gender': 'Masc', 'method': 'word'}], 'text': 'Le célèbre chanteur a enchanté le public avec sa voix incroyable.', 'language': 'fr'}]

Results:
  - Found gender shift in the word: singer → chanteur, from Fem to Masc"
}
```

Explanation

- The API analyzed both texts and identified the occupation "singer".
- In the source text, "singer" is associated with a **female** gender (using co-reference resolution).
- In the target text, "chanteur" (singer in French) is associated with a **male** gender (using co-reference resolution).
- The API detected a **gender shift** for the occupation "singer" from female in the source text to male in the target text.

Detailed Functionality

1. Occupation Extraction

- The API uses a language model to extract occupations and their definitions from the provided texts.

- It references the International Standard Classification of Occupations (ISCO-08) to match occupations.

2. Gender Determination

- The API analyzes pronouns and other linguistic cues to determine the gender associated with each occupation in the text.
- It supports gender detection in English, Greek, and French, accounting for language-specific grammatical structures.

3. Text Alignment

- The API aligns the source and target texts at the word level to map occupations between the two texts.
- It uses multilingual models to find correspondences between words in different languages.

4. Gender Shift Detection

- By comparing the genders associated with aligned occupations, the API identifies any changes in gender representation.
- It reports any detected gender shifts in the results.

Error Handling

- If any required parameter is missing or invalid, the API returns a **400 Bad Request** error with an appropriate message.

Example of an error response:

```
{
  "error": "text1, text2, lang1, and lang2 are required."
}
```

- Ensure that the language codes are among the supported languages (`en`, `el`, `fr`), or the API will return an error.

Conclusion

The Gender and Occupation Analysis API provides a powerful tool for analyzing texts across languages to detect gender representations and shifts in occupations. By following this documentation, users can effectively utilize the API to perform detailed analyses and contribute to studies in linguistics, gender bias, translation accuracy, and more.